

SC_DM270DS DSP Subsystem Specification

Beta 0.1

	Originator	Designer
Name		
Date		
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History

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1 Introduction

Described in this section is the architecture of the DSP Subsystem. The following figure depicts the architecture.

(TBD)

2 DSP Subsystem

The DSP Subsystem contains the cLEAD DSP which is a fully programmable DSP and its associated data and instruction memories as well as peripherals that must be closely coupled with the DSP.

2.1 Components

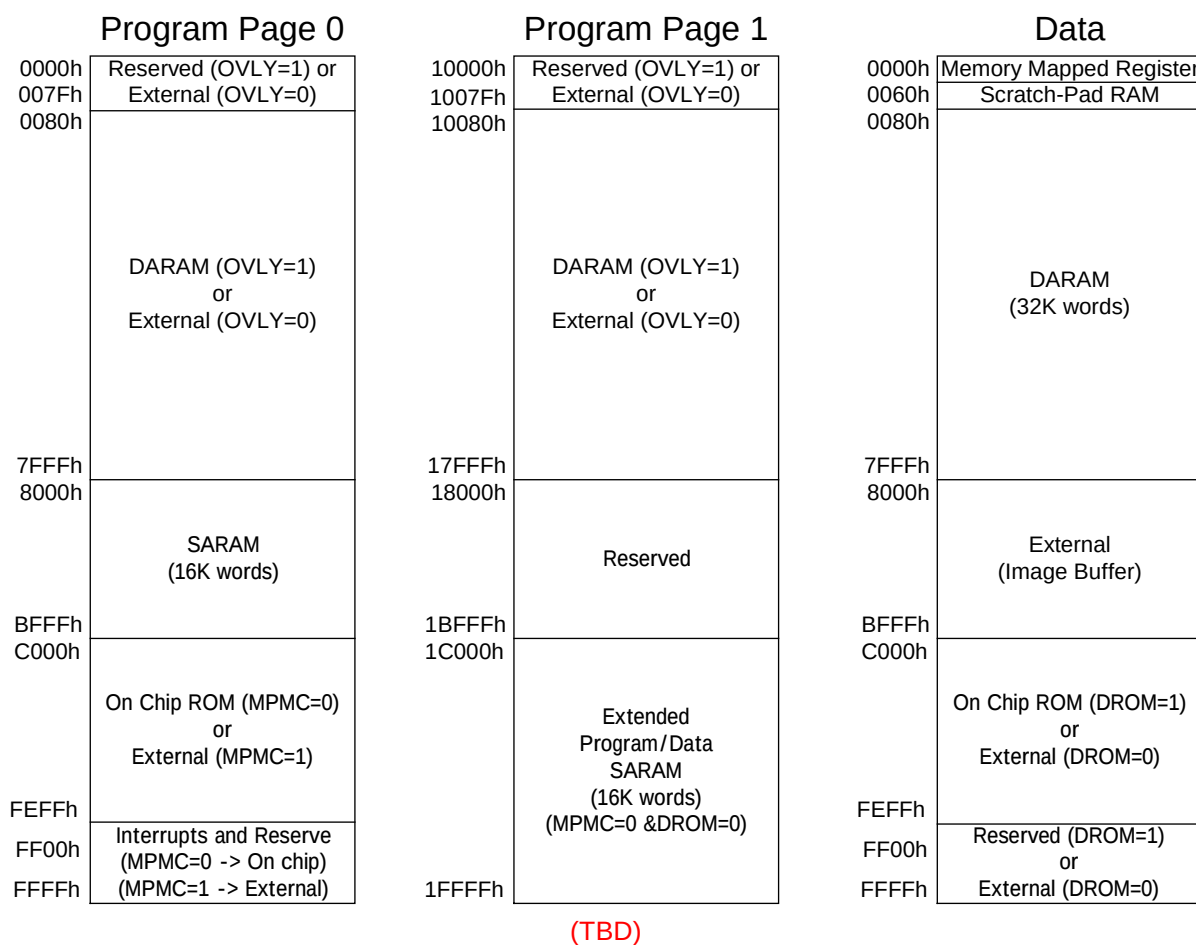
The components comprise the DSP Subsystem are:

- C54x programmable DSP at xxxMHz using the cLEAD2 CPU core.
- 4 k x 32-bit x ?? (??k words) of BRY type ASIC memory providing on-chip zero wait state storage of program and data.
- 4 k x 32-bit x ?? (??k words) of BRY type ASIC memory providing on-chip zero wait state storage of program.
- 4k x 32-bit x 2? (16k? words) of BW type ASIC ROM for zero wait state boot program and standard program data and program storage.
- McBSP programmable serial interface module.
- Programmable general purpose timer.
- Programmable General Purpose IO module providing general purpose interfaces.
- Host Port Interface (HPI) providing direct access to DSP program and data memories.

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2.2 Memory Map

SC_DM270DS DSP memory map is shown next.



The following is a list of the other components and their address relative to the cLEAD Processor.

DSP Subsystem Address Map(TBD)

Address	Function	Access	Commnet
0x0000-0x7FFF	Program/Data Memory	R/W	32k words
0x18000-0x1FFFF	Page 1 Program/data Memory	R/W	32k words
0xC000-0xFFFF	Progam/Data ROM	R	16k words
0x003c-0x003d	GPIO	R/W	2 words

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0x0024-0x0026	Timer	R/W	3 words
0x002c	HPI	R/W	1 word
0x0020-0x0023, 0x0038-0x0039	McBSP0	R/W	6 words plus subbank
0x0040-0x0043, 0x0048-0x0049	McBSP1	R/W	6 words plus subbank
0x0030-0x0035	McBSP2	R/W	6 words plus subbank
0x0054-0x0057	DMA	R/W	4 words plus subbank

2.3 Memory Blocks

The DSP Subsystem contains 5 categories of memories for the cLEAD functions:

- The cLEAD *private* data memories are connected directly to all or individual cLEAD data buses (C, D, E, P) and allow for zero wait state access to all memories.
- The cLEAD (*private*) program memories are connected to the P bus, allow zero wait state access, and are BRY type RAM connected with SARAM wrapper.
- The cLEAD program/data ROMs in the system are mask programmed.

The following table describes the function, size and type of memory:

DSP Subsystem Memories(TBD)

Memory Function	Size	Type	DSP View, Notes	Interfaces
cLEAD Memory Meminst0	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	C, D, E, P Bus M bus – DMA
cLead Memory Meminst1	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	C, D, E, P Bus M bus – DMA
cLead Memory Meminst2	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	C, D, E, P Bus M bus – DMA
cLead Memory Meminst3	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	C, D, E, P Bus M bus – DMA
cLead Memory Meminst4	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	P, E Bus M bus – DMA

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cLead Memory Meminst5	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	P, E Bus M bus – DMA
cLead Memory Meminst6	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	P, E Bus M bus – DMA
cLead Memory Meminst7	4096 X 32	BR04096032160	4096 X 32, API is 32 bits, DSP is 32 bits	P, E Bus M bus – DMA
Data/Program ROM0	4096 X 32	BW04096032160	4096 X 32, No API, DSP is 32 bits.	C, D, P Bus
Data/Program ROM1	4096 X 32	BW04096032160	4096 X 32, No API, DSP is 32 bits.	C, D, P Bus

2.4 Interfaces

There are five interfaces to the DSP Subsystem, these are:

- XIO Interface
- McBSP Serial Interface
- Timer Interface
- GPIO Interface
- Host Processor Interface

The interfaces will be discussed in the following sections. These interfaces allow access to the DSP Subsystem internal memories and allow the DSP Subsystem to control various interfaces and communicate with external entities.

2.4.1 XIO Interface

XIO interface is used to read/write access the memories.

2.4.2 McBSP Interface

This interface allows serial communication between an external entity and the DSP Subsystem. Reference the McBSP (*TMS320C5X Buffered Serial Port Functional Specification*) specification for details for use of this interface.

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2.4.3 Timer interface

This interface provides a strobed output.

2.4.4 General Purpose I/O interface

TBD

2.4.5 Host Processor Interface

C54x DSP provides HPI interface for a host microprocessor/controller to communicate with DSP through sharing DSP's internal memory with the host processor.